Class #14 - Least-Squares Problems

1. Motivation: Suppose $Ax = b$ is inconsistent. Find x such that Ax best approximates b.
   That is, the smaller the distance between b and Ax ($_____$), the better the approximation.

2. The general least-squares problem: Find x that makes
   $||b - Ax||$ as small as possible.

3. Def: If $A \in \mathbf{R^{m \times n}}$ and $b \in \mathbf{R^m}$, a *least-squares solution* of $Ax = b$ is an $\hat{x} \in \mathbf{R^n}$ such that

$$||b - A\hat{x}|| \leq ||b - Ax||$$

   for all $x \in \mathbf{R^n}$.

4. Observation: For any $x \in \mathbf{R^n}$, $Ax \in$ $_____$.
   The closest point to b in Col$A$ = $_____$.
   Note: The equation $Ax = \hat{b}$ is $_____$. Hence, the least-squares solution $_____$.

5. Solution of the General Least-Squares Problem:
   A vector $\hat{x}$ is a least-squares solution of $Ax = b$ if and only if A $_____$ = $_____$.
   Suppose $\hat{x}$ satisfies $A\hat{x} = \hat{b}$.
   Note: $(b - \hat{b}) \in$ $_____$ = (Row(A^T))$^\perp$ = $_____$.
   Hence, $A^T(b - A\hat{x}) =$ $_____$, that is,

$$A^T A\hat{x} = A^T b.$$

   (The above equation is commonly referred to as the *normal equations for* $\hat{x}$.)
   The solution set to the normal equations for $\hat{x}$ is $_____$.

① Attendance

② Par. quiz

③ Class #14 #6 ~

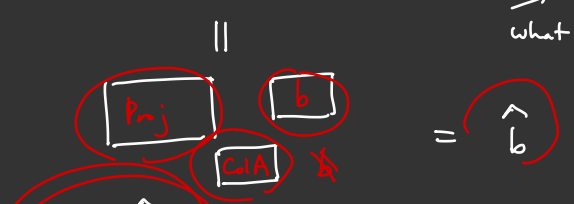
④ Quiz #6 Mon. PP 16-17

⑤ Fri. 11/27 (Early Admission 수시면접 Freshman Interview : Can't use School Buildings for whole day)
→ Video Lecture

Least Squares

① $Ax = b$ is inconsistent

② Look for closest vector to b in Col A.
→ what subspace?



③ $Ax = \hat{b}$ is always Consistent. normal eqn

④ $A\hat{x} = \hat{b}$ $\Leftrightarrow$ $A^TA\hat{x} = A^Tb$

6. Example: Find a least-squares solution of the inconsistent system $Ax = b$ for

$$A = \begin{bmatrix} 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 \end{bmatrix}, \quad b = \begin{bmatrix} -3 \\ -1 \\ 0 \\ 2 \\ 5 \\ 1 \end{bmatrix}.$$

$$A^TA = \begin{bmatrix} 6 & 2 & 2 & 2 \\ 2 & 2 & 0 & 0 \\ 2 & 0 & 2 & 0 \\ 2 & 0 & 0 & 2 \end{bmatrix}, \quad A^Tb = \begin{bmatrix} 4 \\ -4 \\ 2 \\ 6 \end{bmatrix}.$$

$A^TA\,x = A^Tb$

$$\left[\begin{array}{cccc|c} 6 & 2 & 2 & 2 & 4 \\ 2 & 2 & 0 & 0 & -4 \\ 2 & 0 & 2 & 0 & 2 \\ 2 & 0 & 0 & 2 & 6 \end{array}\right]$$

The augmented matrix for the normal equations for $\hat{x}$ is row equivalent to

$$\sim \left[\begin{array}{cccc|c} 1 & 0 & 0 & 1 & 3 \\ 0 & 1 & 0 & -1 & -5 \\ 0 & 0 & 1 & -1 & -2 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right].$$

$t = x_4$ free

$x_1 = 3 - x_4$

$x_2 = -5 + x_4$

$x_3 = -2 + x_4$

$$\hat{x} = \left\{ \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 3-t \\ -5+t \\ -2+t \\ t \end{bmatrix} = \begin{bmatrix} 3 \\ -5 \\ -2 \\ 0 \end{bmatrix} + t \begin{bmatrix} -1 \\ 1 \\ 1 \\ 1 \end{bmatrix} \;\middle|\; t \in \mathbb{R} \right\}$$

The general least-squares solution of $Ax = b$ has the form:

7. Question: When is there only one least-squares solution?

(Square matrix)

A^TA is inv.

8. Theorem: The matrix A^TA is invertible if and only if the columns of A are Linearly indep.

In this case, the equation $Ax = b$ has unique least-squares solution $\hat{x}$, and it is given by

$\hat{x} = (A^TA)^{-1}A^Tb$

Proof:

Claim #1: $\text{Nul } A = \text{Nul}(A^TA)$

(Proof of Claim #1):

① Will show $\text{Nul } A \subseteq \text{Nul}(A^TA)$

Take x such that $Ax = 0$; i.e., $x \in \text{Nul } A$

$\Rightarrow A^TAx = A^T0 = 0$

$\Rightarrow x \in \text{Nul}(A^TA)$

$A = B \Leftrightarrow A \subseteq B$ and $B \subseteq A$

$x \in A \Rightarrow x \in B$

② Will show: $\text{Nul}(A^TA) \subseteq \text{Nul } A$

Take $x \in \text{Nul}(A^TA)$

$\Rightarrow x^TA^TAx = x^T0$

$(Ax)^T(Ax) = 0$

$\|Ax\|^2 = 0$

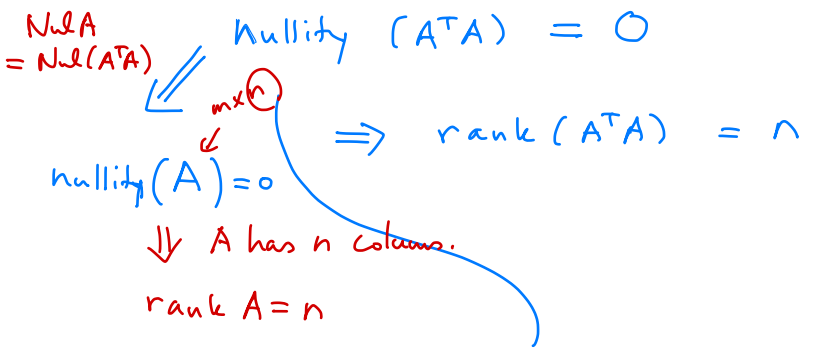
$\Leftrightarrow Ax = 0$

$\therefore x \in \text{Nul } A$

...

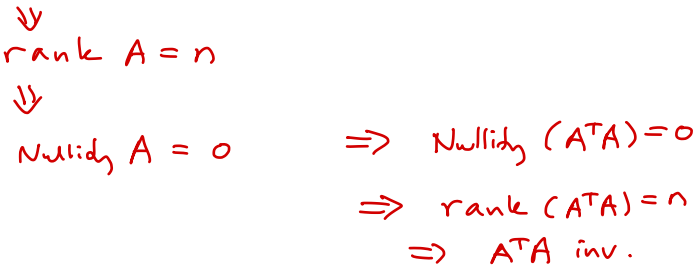
Claim #2: If A is an $m \times n$ matrix such that A^TA is invertible, then the columns of A are linearly independent.

(Proof of Claim #2):



Claim #3: If A is an $m \times n$ matrix whose columns are linearly independent, then A^TA is invertible.

(Proof of Claim #3):

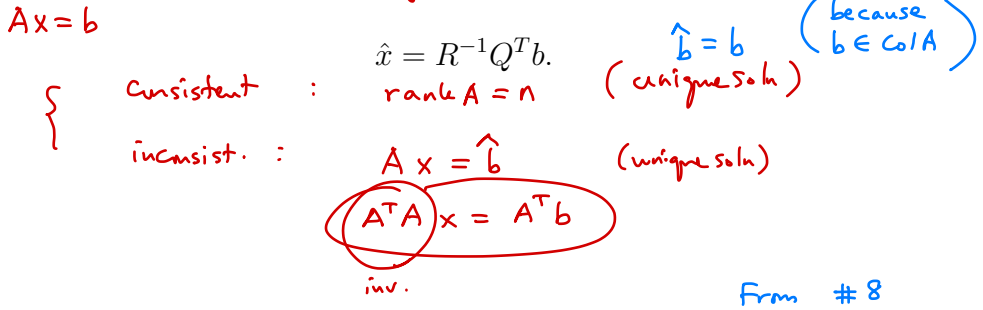


Hence, the 1st statement of the Theorem is true, and the 2nd statement follows.

9. Theorem: Given an $m \times n$ matrix with linearly independent columns (then $A =$ QR), for each $b \in \mathbf{R^n}$, the equation $Ax = b$ has a unique least-squares solution, given by

$$\hat{x} = R^{-1}Q^Tb.$$

Proof:



$A = QR$

From #8

$$\hat{x} = (A^TA)^{-1}A^Tb$$

$$= \left((QR)^T(QR)\right)^{-1}A^Tb$$

$$= (R^TQ^TQR)^{-1}$$

($Q^TQ = I$, $(R^TR)^{-1}$)

$$= R^{-1}(R^T)^{-1}(QR)^Tb$$

$$= R^{-1}(R^T)^{-1}R^TQ^Tb$$

($(R^T)^{-1} = (R^{-1})^T$)

$$= R^{-1}Q^Tb$$

Note: Since R is upper trianguar, it is easier to calculate $R\hat{x} = Q^T b$.

$\hat{x} = R^{-1} Q^T b$ $\quad [R \mid Q^T b]$

$R\hat{x} = Q^T b$ ~

Costs less than computing inverse

$A\hat{x} = \hat{b}$

$\|\hat{b} - b\|$ min.

10. Def: When a least-squares solution $\hat{x}$ is used to produce $A\hat{x}$ as an approximation to b, the distance from b to $A\hat{x}$ ($\|b - \hat{b}\|$) is called the *least-squares error* of this approximation.